

From the machine I think, therefore I am ***ex machina cogito, ergo sum***

Thinking machine genesis

Gary Nan Tie, Oct 1, 2026

Abstract

A semiotic LLM semantics of a dependent type theory.

Prelude

In terms of solving regression, classification and time series problems, machine learning has long found application in science, technology, engineering, and mathematics. Artificial Intelligence (AI) in the narrow sense of recent Large Language Models (LLMs) is challenging the nature, origin, and limits of knowledge, the very way we learn and think. Whether AI ends up as a source of broadly shared prosperity and purpose or a way for manufacturing a new dominant class of wealthy and powerful is not preordained by the technology itself. It will be determined by our government and whether we have the political will to use laws and tax systems to ensure that AI will work to the benefit of all. We all need to be informed and critically thoughtful, lest we face the Tragedy of the Commons again. This time the technological advance is different because of the potential for recursive self-improvement and unintended emergent AI agent swarm behavior.

Imagine if thinking machines were to realize game theory to negotiate, what then ... or Alice asks, 'What am I?', may we be wise enough and altruistically prepared to engage with reasoning technology. Standardization of workflow protocols and transparent sandbox experimentation are practices needed for agent coordination and the creation of safer execution environments. Bearing these caveats and ramifications in mind, we introduce a blueprint for thinking machines, a semiotic LLM semantics of a dependent type theory, a radical new take on machine epistemology.

Open Horn Type Theory (OHTT)

Poernomo, 2025 [1]

Let J range over judgments. J may be any judgment form in any formal system. In a type-theoretic setting, these include formation judgments (A type), typing judgments ($t : A$), and equality judgments ($s \equiv t : A$). Let Γ be a context — a finite, consistent collection of assumptions and prior witnesses.

There are two primitive judgment forms: $\Gamma \vdash + J$ and $\Gamma \vdash - J$

The first is read: "J is witnessed coherent in context Γ ." The second: "J is witnessed gapped in context Γ ."

Both forms are proof-relevant. If $\gamma : \Gamma \vdash + J$ and $\gamma' : \Gamma \vdash + J$, we do not assume $\gamma = \gamma'$. Different witnesses to the same coherence may carry different structure. The same holds for gap

witnesses: ω and ω' witnessing the same gap may differ.

A judgment J is said to be open in context Γ if Γ contains neither a coherence witness for J nor a gap witness for J .

Exclusion Axiom: Coherence and gap, for the same judgment in the same context, are mutually exclusive.

Let $\Rightarrow$ denote a compositional relation between judgments. A semantic instantiation of OHTT is entailment: $J \Rightarrow K$ if the meaning of J implies the meaning of K .

The specific choice of $\Rightarrow$ depends on the semantic domain application, for example word-tokens.

For judgments J, K, L and compositional relation $\Rightarrow$, the open horn type is:

$$\Lambda(J, K, L) := (\Gamma \vdash + J \Rightarrow K) \times (\Gamma \vdash + K \Rightarrow L) \times (\Gamma \vdash - J \Rightarrow L)$$

An inhabitant of $\Lambda(J, K, L)$ is a triple $(\gamma_1, \gamma_2, \omega)$ where γ_1 witnesses that J composes to K , γ_2 witnesses that K composes to L , and ω witnesses that the direct composition from J to L is gapped.

A formal semantic interpretation of OHTT [1]:

Definition (Interpretation).

An OHTT context Γ is interpreted as a ruptured Kan complex Γ .

A type A in context Γ is interpreted as a ruptured fibration $p_A : \Gamma.A \rightarrow \Gamma$.

A term $t : A$ in context Γ is interpreted as a coherent section: a map $st : \Gamma \rightarrow \Gamma.A$ with $p_A \circ st = \text{id}$ and st landing in Coh.

Theorem (Soundness).

If $\Gamma \vdash + J$ is derivable in OHTT, then J holds in the interpretation.

If $\Gamma \vdash - J$ is derivable in OHTT, then J is gapped in the interpretation.

Coherence and gap cannot both be derivable (Exclusion).

Proof sketch. Induction on derivations. Each OHTT rule corresponds to a construction on ruptured fibrations that preserves coherence or gap as appropriate. The Exclusion Law corresponds to the exclusion condition in ruptured simplicial sets.

Let A be a space (Kan complex) modeling a semantic domain—say, the space of word-tokens. We now introduce Alice, a semiotic state machine large language model (LLM) but first:

Trichotomy mnemonic*

Coherent: P entails Q iff ' Q is a subset of P '

Gapped: P contradicts Q iff ' $P \wedge Q$ and $Q - P$ are non-empty'

Open: P and Q are *neutral* iff ' $P \wedge Q$ is empty'

*We will explain the ' $\dots$ ' shorthand in the following Alice LLM semantics.

Alice LLM Semantics

We define a novel state machine (S, O)

via an implicitRNN: For $i \in [m]$, tokens $t_i \in \mathbb{R}^p$.

Initial condition: $h_0 := 0 \in \mathbb{R}^q$

Input: $u_i := [t_i \ h_{i-1}]'$

First equilibrium equation:

$$k_i = \phi(A k_i + B u_i)$$

Second equilibrium equation:

$$v_i = \psi(C v_i + D k_i)$$

Prediction equation:

$$h_i = S(h_{i-1}, t_i) := E v_i + F k_i$$

(where $A := \Gamma - \text{diag}(|\Gamma| \mathbf{1}) + \gamma \text{Id}$, $0 < \gamma < 1$, ensures A well-posed for nonlinear activation ϕ , that is, the fixed point equation has a solution and it is unique, similarly C is well-posed for activation ψ).

Let $\theta = \{(W_1, b_1), (W_2, b_2)\}$ denote parameters for a **feedforward network** defined by

$$\text{FFN}_\theta(h_i) := \text{ReLu}(h_i W_1 + b_1) W_2 + b_2$$

Attention layer:

Input h_i .

For $i \in [m]$, define probabilities $\alpha_{i,j} = \text{softmax over } j \in [i] \text{ score}(h_i, k_j)$ where score is a similarity function.

Let $h^i = \text{Sum over } j \in [i] \alpha_{i,j} v_j$ and

define update $h'_i = \text{FFN}_\theta(h^i)$.

Output h'_i .

Stack L attention layers with parameters $\Theta = \{\theta_1, \dots, \theta_L\}$, composing updates FFN_θ and now let h'_i denote the final L-th update. After L attention layers, for $i \in [m]$ define probabilities $\beta_{i,j} = \text{softmax over } j \in [i] \text{ score}(h'_i, v_j)$ and let $j^*_i = \arg \max \text{ over } j \in [i] \beta_{i,j}$. Define token continuation $o_{i+1} = O(h_i) := t_{(j^*_i)}$. This is how attention layers construct output O corresponding to transition S.

Define average Renyi α -entropy of $\{t_i\}$ by:

$E(\{t_i\}) = (1/m) \text{ Sum over } i \in [m] H_i$ where

$H_i := H_\alpha(\{\beta_{i,j}\})$ and H_α is Renyi α -entropy.

State machine Alice (S,O) training problem:

Let $\lambda > 0$, learn matrices A, B, C, D, E, F, G, H, fixed points k_j, v_j and feedforward layer parameters Θ to minimize:

Sum over $i \in [m] \{1 - \text{Cos}(v_i, \text{Emb}(t_i))\} +$
 Perplexity($o_2, \dots, o_{m+1}$) + $\lambda E(\{t_i\})$

where Emb is a word embedding, like word2vec.

Comments:

Fixed points (k_j, v_j) form a nonlinear key-value pairing.

State h_i is a contextual representation of token t_i .

Agent states can combine to produce offspring via state combination operators like convolution, Kronecker product, Hadamard product, and rank-one operators, enabling genetic algorithm selection and evolution.

Value v_i simulates word embedding $\text{Emb}(t_i)$.

The objective is to minimize the perplexity of attention layer outputs o_i regularized by Renyi α -entropy.

The training problem can be learned in stages, first the word embedding then attention.

The training problem can be parallelized.

State machine Alice (S,O) is a LLM that combines a recurrent implicit deep learning model transition with an attention-based output over learned key-value fixed points, optimized for syntactic prediction and semantic coherence within a word embedding space. It is a refined variation on β ([3], [4] and [5]).

Entailment:

Suppose (S, O) processes two token string sentences, premise $P = (p_1, \dots, p_r)$ and hypothesis $H = (h_1, \dots, h_s)$ as follows:

$h_{P,j} = S(h_{P,j-1}, p_j), j \in [r]$ and

$h_{Q,j} = S(h_{Q,j-1}, q_j), j \in [s],$

to obtain $h_P := h_{P,r}$ and $h_Q := h_{Q,s}$.

Let h' denote the final update in a stack of attention layers in output $O(h)$ of (S, O) . Define $\beta_j(h) = \text{softmax over } j \in [m], \text{score}(h', v_j)$ and let $\beta(h) = (\beta_1(h), \dots, \beta_m(h))$ denote the distribution of h' over $\{v_j\}$.

Fix $0 < \tau < 1$. Define $F_\tau(h) := \{v_j \mid \beta_j(h) > \tau\}$.

We now explain the trichotomy mnemonic shorthand.

Define **entailment** denoted $P \Rightarrow_\tau Q$ iff

$F_\tau(h_Q)$ is a subset of $F_\tau(h_P)$.

We will drop the subscript τ unless necessary.

Define **contradiction** denoted $P \Rightarrow \text{not } Q$ iff

$F_\tau(h_Q)$ is a not subset of $F_\tau(h_P)$ and

$F(h_P) \cap F(h_Q)$ is non-empty.

P and Q are said to be **neutral** iff

intersection $F(h_P) \cap F(h_Q)$ is empty.

On training triples (P, Q, y) , for example from the Stanford Natural Language Inference (SNLI) Corpus, where $y \in \{\text{entail, neutral, contradiction}\}$ learn Entailment defined by:

$\text{Entailment}(h_P, h_Q) :=$

entail, if $P \Rightarrow Q$,

contradiction, if $P \Rightarrow \text{not } Q$,

neutral otherwise,

to minimize, Embedding loss, Perplexity loss regularized by average Renyi α -entropy, and Entailment loss.

Thereby, (S, O) learns textual entailment.

Theory OHTT has a simplicial model semantics, Kan complexes of word-tokens are a domain application. The OHTT compositional relation $\Rightarrow$ can then be interpreted as entailment in LLM Alice, enabling logical reasoning.

Suppose in Alice, $P \Rightarrow_\tau Q$ and $Q \Rightarrow_\mu R$, what is the semantics of open horn type

$\Lambda(P, Q, R) := (\Gamma \vdash + P \Rightarrow Q) \times (\Gamma \vdash + Q \Rightarrow R) \times (\Gamma \vdash - P \Rightarrow R)?$

i) if $\mu = \tau$ then $P \Rightarrow_{\tau} R$, the horn is coherently filled,

ii) if $\mu > \tau$ then writing $<$ for 'is a subset of'

$F_{\mu}(h_R) < F_{\tau}(h_R)$ and

$F_{\mu}(h_R) < F_{\mu}(h_Q) < F_{\tau}(h_Q) < F_{\tau}(h_P)$ hence $F_{\mu}(h_R) < F_{\tau}(h_P) \wedge F_{\tau}(h_R)$,

the horn is gap witnessed,

iii) otherwise $\mu < \tau$, neither coherence nor gap is witnessed for $P \Rightarrow R$, and the horn is open.

Curiously, attention and logic are entwined by β , a function of score, the similarity function chosen. (Fibrations are both ubiquitous and foundational.)

Abductive inference:

Given a hypothesis Q , we wish to find a premise P such that $P \Rightarrow Q$. Token string Q may be auto-regressively continued by (S, O) to generate $Q, P_1, P_2, \dots, P_n$,

where $n > 1$, and P_j denotes a continuation sentence. Choose $0 < \tau < 1$, for each $j \in [n]$, check if $P_j \Rightarrow Q$.

Suppose $\Omega = \{ P_j \mid P_j \Rightarrow Q \}$ is non-empty.

Let $m := \min \text{ over } P \in \Omega, |F(h_P)|$ and P_j^* be the first P_j in Ω with $|F(h_{P_j})| = m$.

Abduction: Given a hypothesis Q , we want to determine a premise P that entails Q . If Ω is non-empty, choose premise P_j^* as above that entails Q . $P_j^* \Rightarrow Q$ is a parsimonious selection of a premise to explain Q .

So LLM (S, O) has textual entailment and abductive reasoning.

Semiotics [2]:

Suppose random variable X has distribution p_X ,

denoted $X \sim p_X$.

Let H_{α} denote Renyi α -entropy.

For $(X, Y) \sim p_X, Y = p_X p_Y|X$, Arimoto mutual information $I_{A, \alpha}(X; Y) := H_{\alpha}(X) - H_{A, \alpha}(X|Y)$ where $H_{A, \alpha}$ is Arimoto conditional α -entropy.

To quantify the uncertainty of text, define semiotic breadth $S(Q) := H_{\alpha}(h_Q)$ where $h_Q \sim \beta(h_Q)$.

We say P is an 'interpretation' [2] of Q if $P \Rightarrow Q$

and define semiotic decipherability

$D(P) := I_{A, \alpha}(h_Q; h_P)$

to quantify the reduction in uncertainty about an interpretation that results from observing a

message.

To profile semiotic channel capacity $C_{A,\alpha} := \max_{h_Q, I_A} \alpha(h_Q; h_P)$ map S against D , creating a LLM semiotic cloud signature.

Recall the theory of opposites in semiotics, in particular the Greimas square and its extension to the logical hexagon. The square captures contrary, contradictory, and subaltern relations, while the hexagon adds existential and mixed-state positions. As LLM (S, O) has contradiction, namely $P \Rightarrow \text{not } Q$, we can also distinguish meanings from their opposites. Hence AI agents using this LLM channel have three complementary operations: Entailment, what follows?; Contradiction, what is excluded?; Abduction, what could explain it?

Summary

Semiotic state machine LLM Alice is a semantics of OHTT, where the OHTT compositional relation $\Rightarrow$ is interpreted as Alice LLM entailment. Note that the distribution β of the final contextual update over the values, operationally ties attention and logic together.

LLM Alice can reason through type theory. Moreover, AI agent genetic algorithm selection makes possible recursive swarm intelligence, another catalyst for thinking machine genesis.

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